Matlab 快速入门

大学生数学建模基础课程

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Notes

简介

简史

MATLAB (矩阵实验室) 是 **MAT**rix **LAB**oratory 的缩写;

最初由美国的 Clever Moler 教授于 1980 年开发, 初衷是为了解决 "线性代数" 课程的矩阵运算问题;

是一款由 MathWorks 公司 (1984 年成立) 出品的数学软件.

特性

MATLAB 是一种用于算法开发, 数据可视化, 数据分析以及数值计算的高级技术计算语言和交互式环境.

MATLAB 可用来创建用户界面及调用其它语言编写的程序.

MATLAB 中包含众多的附加工具箱, 适合不同领域的应用.

Notes

窗口介绍

命令窗口

脚本窗口

图形窗口

Notes

获得帮助

帮助文档

如果你知道一个函数名, 想了解它的用法, 你可以用 'help' 命令得到它的帮助文档:

```
>> help functionname
```

如果你了解含某个关键词的函数, 你可以用 'lookfor' 命令得到相关的函数:

```
>> lookfor keyword
```

网络资源

Mathworks 文件交流中心: ▸ Mathworks

Github 代码托管网站: ▸ Github

Notes

实数, 复数, 行向量, 列向量, 矩阵的赋值

```
Command Window
>> x = 5
  x =
       5
>> x = [1 2 3]
  x =
       1     2     3
>> x = [1;2;3]
  x =
       1
       2
       3
>> clc
```


Notes

实数, 复数, 行向量, 列向量, 矩阵的赋值

```
Command Window
>> x = [1 2 3; 4 5 6; 7 8 9]
  x =
       1     2     3
       4     5     6
       7     8     9
>> y = [1 2 3
        4 5 6]
  y =
       1     2     3
       4     5     6
>>
```


Notes

向量的一般赋值方法

```
Command Window
>> x = [0:2]
  x =
    0.00    1.00    2.00
>> x = [0:2]'
  x =
    0.00
    1.00
    2.00
>> x = [0:0.5:2]
  x =
    0.00    0.50    1.00    1.50    2.00
>> x = linspace(0, 2, 5)
  x =
    0.00    0.50    1.00    1.50    2.00
>>
```


Notes

常用矩阵

```
Command Window
>> x = zeros(2,3)
  x =
    0.00    0.00    0.00
    0.00    0.00    0.00
>> y = ones(2)
  x =
    1.00    1.00
    1.00    1.00
>> x = eye(2)
  x =
    1.00    0.00
    0.00    1.00
>> z = rand(1,2)
  z =
    0.23    0.96
>>
```


Notes

固定变量

```
Command Window
>> pi
ans =
    3.1416
>> z = i
z =
   0.00 + 1.00i
>> x = 1/0
x =
   Inf
>> 0/0
ans =
   NaN
>>
```


Notes

矩阵运算和数组运算

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> B = [1 3 5; 6 9 0; 2 4 6];
>> C = A + B
C =
     2     5     8
    10    14     6
     9    12    15
>> D = A - B
D =
     0    -1    -2
    -2    -4     6
     5     4     3
>> clc
```


Notes

矩阵运算和数组运算

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> B = [1 3 5; 6 9 0; 2 4 6];
>> E = A * B
E =
    19    33    23
    46    81    56
    73   129    89
>> F = A.* B
F =
     1     6    15
    24    45     0
    14    32    54
>> clc
```


Notes

矩阵运算和数组运算

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> B = [1 3 5; 6 9 0; 2 4 6];
>> G = A / B
G =
         0         0    0.50
   -3.00    0.00    3.50
   -6.00    0.00    6.50
>> H = A./ B
H =
    1.00    0.67    0.60
    0.67    0.56     inf
    3.50    2.00    1.50
>> clc
```


Notes

矩阵运算和数组运算

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> B = [1 3 5; 6 9 0; 2 4 6];
>> I = A ^ 2

  I =

      30   36   42
      66   81   96
     102  126  150

>> J = A.^ 2

  J =

       1    4    9
      16   25   36
      49   64   81

>> clc
```


Notes

数组和数组行列块操作: 取值

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> x = A(1, 3)

  x =

       3

>> y = A(2, :)

  y =

       4    5    6

>> z = A(1:2, 1:3)

  z =

       1    2    3
       4    5    6

>>
```


Notes

数组和数组行列块操作: 赋值

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> A(1, 3) = 0

  A =

       1    2    0
       4    5    6
       7    8    9

>> A(2, :) = [6 5 4]

  A =

       1    2    0
       6    5    4
       7    8    9

>> A(1:2, 1:2) = [-1 -2; -3 -4]

  A =

      -1   -2    0
      -3   -4    4
       7    8    9
```


Notes

比较和逻辑运算

```
Command Window
>> x = [1  2  3  4  5  6  7  8  9];
>> y = [1  4  3  8  6  5  7  2  9];
>> eq = (x==y)

  eq =

       1    0    1    0    0    0    1    0    1

>> xy = (x>5)&(y<7)

  xy =

       0    0    0    0    0    1    0    1    0

>> xoy = (x>5)|(y<7)

  xoy =

       1    1    1    0    1    1    1    1    1

>> xory = xor(x>5,y<7)

  xory =

       1    1    1    0    1    0    1    0    1

>>
```


Notes

比较和逻辑运算

```
Command Window
>> x = [1 -2  3 -4  5 -6  7 -8  9];
>> x(x<0) = 0

  x =

       1    0    3    0    5    0    7    0    9

>> y = [1  2  3;-4  5  6;  7  8  9];
>> y(y(:,1)<0,:) = 0

  y =

       1    2    3

       0    0    0

       7    8    9

>>
```


Notes

数组操作函数: flipud, fliplr, rot90

```
Command Window
>> A = [1 2 3; 4 5 6; 7 8 9];
>> B = flipud(A)

  B =

       7    8    9
       4    5    6
       1    2    3

>> C = rot90(A)

  C =

       3    6    9
       2    5    8
       1    4    7

>>
```


Notes

数组操作函数: sum

```
Command Window
>> A = [1 2 3];
>> sum(A)

  ans =

       6

>> B = [1 2 3; 4 5 6; 7 8 9];
>> sum(B)

  ans =

      12   15   18

>> sum(B,2)

  ans =

       6
      15
      25

>>
```


Notes

数组操作函数: max, min

```
Command Window
>> A = [1 2 3];
>> max(A)

  ans =

       3

>> max(A,2)

  ans =

       2    2    3

>> B = [1 3 9; 4 8 6];
>> max(B)

  ans =

       4    8    9

>> max(B, [], 2)

  ans =

       9
       8
```


Notes

常用数学函数: sin, cos, tan, cot, asin, acos, atan, acot

```
Command Window
>> x = 0:pi/6:pi;
  x =
     0.00  0.52  1.05  1.57  2.09  2.62  3.14
>> y = sin(x)
  y =
     0.00  0.50  0.87  1.00  0.87  0.50  0.00
>> z = asin(y)
  z =
     0.00  0.52  1.05  1.57  2.09  2.62  3.14
>>
```


Notes

常用数学函数: abs, sqrt

```
Command Window
>> x = [-4 9 -16 25];
  x =
     -4    9  -16   25
>> y = abs(x)
  y =
      4    9   16   25
>> z = sqrt(y)
  z =
      2    3    4    5
>>
```


Notes

常用数学函数: ceil, fix, floor, round

```
Command Window
>> x = [-1.6 -0.2 1.2 0.6];
>> y = ceil(x)
  y =
     -1     0     2     1
>> z = floor(x)
  z =
     -2    -1     1     0
>> g = fix(x)
  g =
     -1     0     1     0
>> f = round(x)
  f =
     -2     0     1     1
>>
```


Notes

基本语句

基本语句

```
for .. end
if .. else .. end
while .. end
switch .. case .. end
```

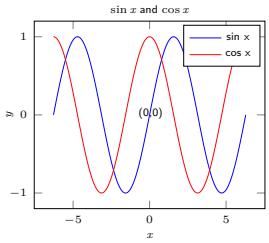
举例: 求 1-10 以内的奇数和

```
% sum of the odd numbers between 1 and 10
x = 0;
for i = 1:10
    if mod(i,2)
       x= x + i;
    end
end
```


Notes

简单作图

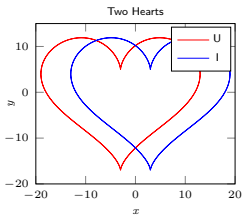
```
1  x = -2*pi:0.1:2*pi;
2  y1 = sin(x);
3  y2 = cos(x);
4  plot(x, y1, '-b');
5  hold on
6  plot(x, y2, '-r');
7  xlabel('x')
8  ylabel('y')
9  text(0,0, '(0,0)')
10 legend('sin x', 'cos x')
11 title('sin x and cos x')
```



Notes

简单作图

```
1  t = 0:pi/180:4*pi;
2  x = 16*sin(t).^3;
3  y = 13*cos(t)-5*cos(2*t)...
4      -2*cos(3*t)-cos(4*t);
5  plot(x-3,y,'-r', x+3,y,'-b');
6  xlabel('x');
7  ylabel('y');
8  axis([-20, 20, -20, 15]);
9  title('Two Heart')
10 legend('U', 'I')
```



Notes

简单作图

曲线图 plot: plot(x,y); plot(x,y,s), plot(x1,y1,s1,x2,y2,s2,...)

```
1  b  blue        .  point          -   solid
2  g  green       o  circle         :   dotted
3  r  red         x  x-mark         -.  dashdot
4  c  cyan        +  plus           --  dashed
5  m  magenta     *  star         (none)  no line
6  y  yellow      s  square
7  k  black       d  diamond
8  w  white       v  triangle (down)
9                 ^  triangle (up)
10                <  triangle (left)
11                >  triangle (right)
12                p  pentagram
13                h  hexagram
```


Notes

简单控制语句

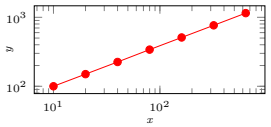
title(图形名称)
xlabel(x 轴说明); ylabel(y 轴说明)
text(x,y, 图形说明)
legend(图例 1, 图例 2, ...)
grid on / grid off / grid minor
axis([xmin xmax ymin ymax]), xlim([xmin, xmax])

Notes

其它坐标系

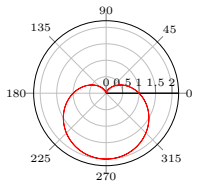
对数坐标: loglog, semilogx

```
1  x = 10*2.^[0:6];
2  y = [100 150 225 340 ...
3       510 765 1150];
4  loglog(x,y,'.-r')
5  xlim([0.5e1,0.8e3])
6  ylim([0.8e2,1.4e3])
7  xlabel('x'); ylabel('y')
```



极坐标: polar

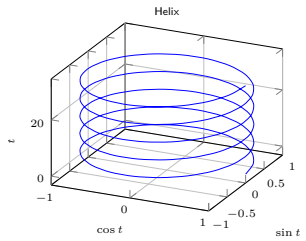
```
1  theta=0:pi/180:4*pi;
2  r=1-sin(theta);
3  polar(theta,r,'-r');
```



Notes

三维曲线图

```
1  t=0:pi/50:10*pi;
2  x = sin(t);
3  y = cos(t);
4  z = t;
5  plot3(x,y,z)
6  title('Helix')
7  xlabel('sin t')
8  ylabel('cos t')
9  zlabel('t')
10 grid on
```



Notes

三维曲面图：补充函数 meshgrid

```
Command Window
>> [x, y] = meshgrid(1:3, 1:3)
  x =
       1     2     3
       1     2     3
       1     2     3
  y =
       1     1     1
       2     2     2
       3     3     3
>> rsq = (x-2).^2 + (x-2).^2
  rsq =
       2     1     2
       1     0     1
       2     1     2
>> r = sqrt(rsq)
  r =
    1.4142    1.0000    1.4142
    1.0000         0    1.0000
    1.4142    1.0000    1.4142
```

(1,1)	(2,1)	(3,1)
(1,2)	(2,2)	(3,2)
(1,3)	(2,3)	(3,3)

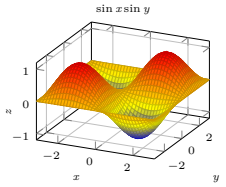
2	1	2
1	0	1
2	1	2

$\sqrt{2}$	1	$\sqrt{2}$
1	0	1
$\sqrt{2}$	1	$\sqrt{2}$

Notes

三维曲面图

```
1  [x,y] = meshgrid(-pi:0.1:pi);
2  z = sin(x).*cos(y);
3  mesh(x,y,z) % meshc(x,y,z)
4  surf(x,y,z) % surfc(x,y,z)
5  xlabel('x')
6  ylabel('y')
7  zlabel('z')
8  title('sin x sin y')
```



Notes

M 函数格式

M 函数格式

```
function [output 1, ..] = functionname(input1, ..)
% comment of this function

MatLab command 1;
MatLab command 2;
```

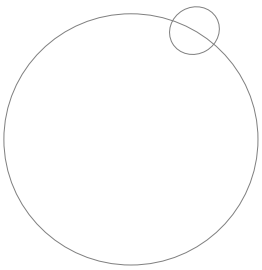
举例: 求矩形面积

```
function area = rectarea(L, W)
% rectarea  Area of a rectangle
%
%   rectarea(l, w) calculate the area of a rectangle
%   with a length of L and a width of W

area = L .* W
```


Notes

多体问题



考虑多个天体的系统 (比如“日地月”三天体系统), 求各个天体的运动规律.

天体间的距离远大于天体的尺寸, 所有天体都视为质点.

每个天体有固定质量, 并给出初始位置和初始速度.

任意两天体之间只有万有引力的作用.

$$\mathbf{F_{ij}} = \frac{Gm_i m_j}{r_{ij}^2}\hat{\mathbf{r}}_{\mathbf{ij}}$$

Notes

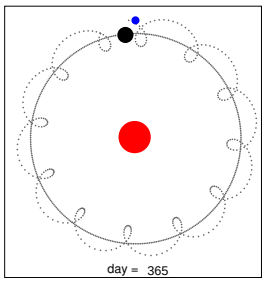
多体问题模拟程序

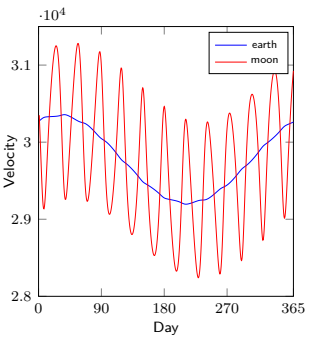
Matlab 程序: main.m

```
01 G = 6.67e-11;  dt = 24*3600;  N = 3;
02 M = [sun.mass    ; earth.mass    ; moon.mass    ];% N X 1
03 R = [sun.position; earth.position; moon.position];% N X 3
04 V = [sun.velocity; earth.velocity; moon.velocity];% N X 3
05 for t = 1:365
06     F = zeros(N,3);                % F(i,:) = [fx, fy, fz]
07     for i = 1 : N
08         mi = M(i); ri = R(i,:);    % 第i个天体的质量和位置
09         for j = (i+1):N;
10             mj = M(j); rj = R(j,:);% 第j个天体的质量和位置
11             rij =  rj - ri;
12             fij = G*mi*mj./(norm(rij).^3).*rij;% 万有引力
13             F([i,j],:) = F([i,j],:) + [fij; -fij];
14         end
15     end
16     V = V + F./repmat(M,1,3)*dt;   % v(t+dt)=v(t)+a(t+dt)dt
17     R = R + V*dt;                  % r(t+dt)=r(t)+v(t+dt)dt
18 end
```

Notes

多体问题模拟结果





Notes

Thank You!!!

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